

ALEXIS MARCHAND

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EMPLOYMENT

from 9/2025 to 9/2027	IM PAN, Warszawa (Poland) Adiunkt (Assistant Professor) Research project: <i>Analysis on groups</i> (NCN Maestro 13) Mentor: PIOTR NOWAK
from 5/2026 to 7/2026	Universität Regensburg / Humboldt-Stiftung (Germany) Humboldt Research Fellow
from 5/2027 to 7/2027	Research project: <i>Invariants of groups and non-positive curvature</i> Mentor: CLARA LÖH
from 9/2024 to 9/2025	Kyōto University / JSPS (Japan) JSPS Postdoctoral Fellow Research project: <i>Stable commutator length in finitely presented groups</i> Mentor: KŌJI FUJIWARA

EDUCATION

from 10/2021 to 10/2024	University of Cambridge (U.K.) • Ph.D. in Pure Mathematics Thesis: <i>Computations and lower bounds for scl and the relative Gromov seminorm</i> Defended on: 15/7/2024 Funding: EPSRC PhD studentship Supervisor: HENRY WILTON
from 9/2017 to 8/2021	École Normale Supérieure de Lyon (France) • Dipl. ENS Funding: ENS studentship awarded by national competition (rank: 16)
from 9/2020 to 7/2021	Université Grenoble Alpes (France) • M.Sc. in Mathematics Mention Très Bien Thesis: <i>Automorphismes extérieurs de produits libres : revêtements abéliens caractéristiques et représentations libres</i> Supervisor: FRANÇOIS DAHMANI
from 10/2019 to 6/2020	University of Cambridge (U.K.) • M.A.St. in Mathematics Pass with Merit Essay: <i>Rigidity theorems for hyperbolic groups</i> Supervisor: HENRY WILTON
from 9/2017 to 7/2019	ÉNS de Lyon / Université Lyon 1 (France) • B.Sc. in Mathematics First two years of ENS curriculum in mathematics (L3+M1)

PREPRINTS

- [1] Alexis Marchand, Piotr Nowak, On vanishing and Hausdorffness of asymptotic cohomology.
arXiv:2609.29872

PUBLICATIONS (PEER-REVIEWED)

- [2] Alexis Marchand, From letter-quasimorphisms to angle structures and spectral gaps for scl, **Israel J. Math.** **273** (2026), no. 2, 937-967. DOI:10.1007/s11856-026-2898-3 • arXiv:2402.13856
- [3] Alexis Marchand, Bavard duality for the relative Gromov seminorm, **Enseign. Math.** **72** (2026), no. 1-2, 1-46. DOI:10.4171/lem/1087 • arXiv:2305.02149
- [4] Alexis Marchand, Isometric embeddings of surfaces for scl, **Groups Geom. Dyn.** **19** (2025), no. 4, 1445-1478. DOI:10.4171/ggd/845 • arXiv:2302.04133
- [5] Rémi Coulon and Alexis Marchand, Growth and co-growth in negatively curved groups, in Dahmani, Erschler, Horbez, Wise (eds.), Groups acting on fractals, hyperbolicity and auto-similarity, **Panoramas et Synthèses** **63** (2025).
- [6] Alexis Marchand, Free representations of outer automorphism groups of free products via characteristic abelian coverings, **J. Group Theory** **26** (2023), no. 2, 399-420.
DOI:10.1515/jgth-2021-0154 • arXiv:2107.11230

EXPOSITORY ARTICLES

- [7] Alexis Marchand, Makers of patterns: from Escher to Coxeter, **Eureka** (2020).
- [8] Alexis Marchand, Géométrie des groupes et courbure négative, **Journal de Mathématiques des Élèves de l'ENS de Lyon** (2020).

GRANTS & FUNDING

from 5/2026 to 7/2027	Humboldt Stiftung • Humboldt fellowship's subsidy towards research costs Allocated amount: EUR 4,800 Topic: <i>Invariants of groups and non-positive curvature</i>
from 9/2024 to 9/2025	JSPS • Fellow's Research Support Allowance Allocated amount: JPY 840,000 Topic: <i>Stable commutator length in finitely presented groups</i>
6/2025	RIMS, Kyōto University • RIMS Workshop (type A) Allocated amount: JPY 350,000 Topic: <i>Spectrum problems for scl via S-machines</i>

AWARDS

- 3/2023 | **Smith-Knight & Rayleigh-Knight prize** • University of Cambridge
Grade 3 (GBP 300), for the essay *Isometric embeddings of surfaces for scl*

RESEARCH VISITS

- 3/2026 | **Kyōto University** (Japan) • One-week visit • invited by RYŌKICHI TANAKA
- 11/2025 | **INI, Cambridge** (U.K.) • Three-week visit
Programme: *Operators, graphs, groups* (participant)
- 6/2025 | **RIMS, Kyōto University** (Japan) • Two-week small-group workshop
Programme: *Spectrum problems for scl via S-machines* (co-organiser)
- 6/2025 | **KIAS, Seoul** (South Korea) • One-week visit • invited by SANG-HYUN KIM
- 4/2025 | **University of Hawai'i at Mānoa** (U.S.) • Ten-day visit • invited by YASH LODHA
- 4/2022 | **IHP, Paris** (France) • One two-week visit and one three-week visit
and 6/2022 | Programme: *Groups acting on fractals* (participant)
- from 6/2019 | **IMPA, Rio de Janeiro** (Brazil) • Two-month M.Sc. internship
to 8/2019 | Topic: *Topology of complex affine varieties* • Supervisor: HOSSEIN MOVASATI
- from 5/2018 | **KIT, Karlsruhe** (Germany) • Six-week B.Sc. internship
to 6/2018 | Topic: *Geometry of Coxeter groups* • Supervisor: CATERINA CAMPAGNOLO

HIGHER EDUCATION TEACHING EXPERIENCE

- from 10/2026 | **Hyperbolic geometry** • Uniwersytet Warszawski
to 1/2027 | Graduate course • 60 contact hours • in English
- from 1/2023 | **Geometry** • Trinity College & Peterhouse College, Cambridge
to 5/2024 | Part IB supervisions • 34 contact hours • in English
- from 1/2024 | **Geometric Group Theory** • University of Cambridge
to 3/2024 | Part III examples classes • 8 contact hours • in English
- from 10/2022 | **Analysis & Topology** • Trinity College & Peterhouse College, Cambridge
to 1/2024 | Part IB supervisions • 46 contact hours • in English
- from 12/2022 | **Admissions interviews** • Christ's College, Cambridge
to 12/2023 | Co-interviewer for undergraduate admissions in mathematics • in English
- from 1/2023 | **Group Cohomology** • University of Cambridge
to 5/2023 | Part III drop-in sessions • 4 contact hours • in English
- from 1/2022 | **Riemann Surfaces** • Trinity College & Peterhouse College, Cambridge
to 5/2022 | Part II supervisions • 31 contact hours • in English
- from 10/2021 | **Linear Algebra** • Trinity College, Cambridge
to 1/2022 | Part IB supervisions • 16 contact hours • in English
- from 10/2021 | **Mapping Class Groups** • University of Cambridge
to 1/2022 | Part III drop-in sessions • 4 contact hours • in English
- from 9/2018 | **Colles de Mathématiques** • Lycée Aux Lazaristes, Lyon
to 5/2019 | Oral examinations in mathematics for first-year students in a preparatory class • 44 contact hours • in French

VOLUNTEER TEACHING

- 3/2025 | **JSPS science dialogue** • Koshi High School, Fukui
Lecture in English and Japanese for a high school science club
- from 10/2023 | **High school project supervision** • Rusanivskyi Lyceum, Kyiv (online)
to 4/2024 | Student: OLHA MYKHAILYK • Topic: *Inversions and Möbius transformations*
Project received the second prize at the city stage in the Junior Academy of Sciences of Ukraine research competition
- from 10/2019 | **Tutoring** • Coleridge Community College & St Faith's School, Cambridge
to 3/2020 | Volunteer tutoring in mathematics in secondary schools, under the *Stimulus* programme
- from 9/2017 | **French as a foreign language** • ItinérENS, Lyon
to 6/2020 | French classes for refugees • more than 100 contact hours

SERVICE

Refereeing and quick opinions for *Algebr. Geom. Topol.*, *Enseign. Math.*, *KIAS Springer Ser. Math.*, *Trans. Amer. Math. Soc.*

Reviewing for *zbMATH Open*

- from 9/2025 | **Analytic & geometric group theory seminar** • IM PAN, Warszawa
to 9/2027 | Co-organiser
- 9/2025 | **World of GroupCraft V** • Online
Co-organiser
- 6/2025 | **RIMS Workshop** • RIMS, Kyōto University
Topic: *Spectrum problems for scl via S-machines* • Co-organiser
- from 2/2025 | **Reading seminar** • Kyōto University
to 4/2025 | Topic: L^2 -Betti numbers • Organiser
- from 9/2023 | **Junior geometry seminar** • University of Cambridge
to 8/2024 | Co-organiser
- from 8/2023 | **MCR committee** • Trinity College, Cambridge
to 5/2024 | MCR (Middle Combination Room) committee member in charge of: the MCR's website, mailing list, elections, lunchtime seminars, etc.
- from 10/2022 | **Reading seminar** • University of Cambridge
to 12/2022 | Topic: *Generalisations of hyperbolic groups* • Organiser

INVITED TALKS (SELECTION)

- 10/2026 | **Università di Pisa / Scuola Normale Superiore**
LiT IV Landing in Topology
- 6/2026 | **ICMAT, Madrid**
Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future
- 6/2026 | **Université Bourgogne Europe**
Séminaire géométrie, algèbre, dynamique et topologie

6/2026	Université de Lorraine Séminaire de géométrie différentielle
11/2025	Newcastle University Algebra & geometry seminar
11/2025	University of Cambridge Geometric group theory seminar
7/2025	University of Tōkyō Tuesday seminar on topology
7/2025	Tōhoku University Geometry seminar
6/2025	KIAS, Seoul Topology seminar
4/2025	University of Hawai'i at Mānoa Mathematics department colloquium
11/2024	Kyōto University First MMS workshop for young researchers
3/2024	University of Oxford Geometric group theory seminar
4/2023	KU Leuven Methusalem junior seminar
3/2023	ETH Zürich Geometry graduate colloquium

LANGUAGES

- French • Native language
- English • Fluent (C2)
Passed the **CAE (Cambridge)**
- Japanese • Advanced (B2)
Passed the **JLPT N2**
- German • Advanced (B2)
- Portuguese • Intermediate (B1)
- Polish • Intermediate (A2/B1)
- Russian • Intermediate (A2/B1)

September 25, 2026